

Estimating Vulnerability-Related Police Demand in Bradford

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The Changing Nature of Policing

The role of the police has evolved over time, and they are now being viewed as the primary public service in many situations. This shift in perception is a result of their versatile functions and responsibilities. Today, the police are increasingly tasked with managing and preventing harm to and from vulnerable populations. To best inform this expanded scope of duties we sought to explore the changing nature and temporal and spatial dependencies related to both current and historical police related demand across the Bradford region. By recognising and analysing these patterns, we hope to inform police and their partners of the best ways to allocate resources and effectively respond to the needs of the communities they serve.

The Data

Police incident data, is when people called West Yorkshire Police (WYP) for police service. In this study we analysed incidents graded emergency or priority and physically attended by police officers.

The study period was April 2017 to March 2022. WYP received 330k calls for service. There are 89 types, 21 subtypes and 39 qualifiers.

Estimating Vulnerability Related Demand

To try and understand which incidents might be indicative of vulnerability we asked a panel of 3 experts to examine a range of markers collected for each incident: "Would you consider this type/subtype/qualifier to be indicative of a vulnerability?"

Consensus Mechanism is defined such that: Incidents classified as vulnerability-related, if the majority of experts consider type, subtype or qualifier(s) to be vulnerability-related.

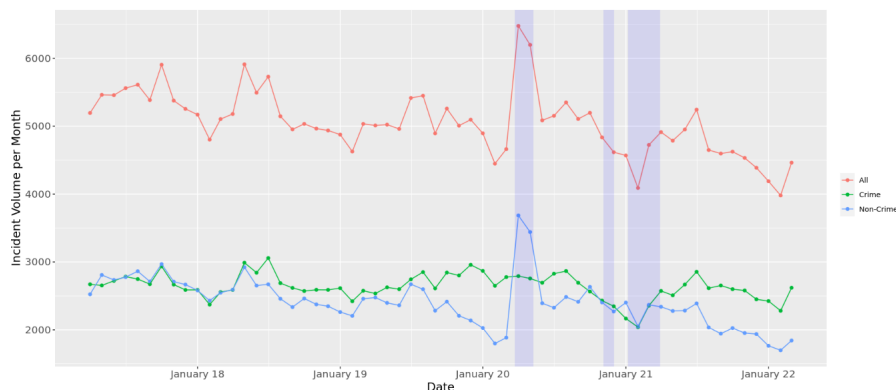


Figure 1: Incident volume plot for attended incidents graded emergency and priority in the five year study period. Trend for all incidents (red), crime related (green) and non crime related (blue). Blue shaded region denotes national lockdown restrictions.

Type	Sub Type	Qualifiers	Vulnerability Related
Sexual Offence	Victim aged 16+	Mental Health, Domestic	Yes
Burglary - residential	Niche created occurrence	Domestic abuse	Yes
Burglary - residential	Niche created occurrence	No app ho qualifier	No

Figure 2: Table illustrating how the Consensus Mechanism Coding works, when determining whether or not an incident is vulnerability-related.

Quantifying Vulnerability-Related Demand

There are two measures of demand: The volume of calls recieved over a given time period and duration of time associated with responding to each of those calls. We combine these measures to calculate accumulated response time - measure of the total amount of time spent responding to particular incidents in each neighbourhood. Using Consensus Mechanism to identify Vulnerability related Incidents.

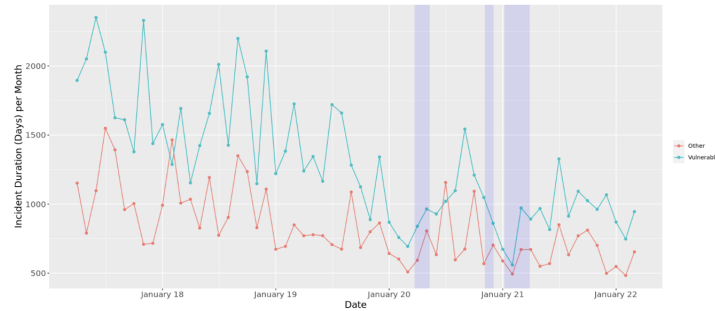


Figure 3: Temporal trends for Vulnerability-Related incident durations across the five year study period, considering attended incidents of emergency and priority grading. Blue shaded region denotes national lockdown restrictions.

Spatial Distribution of Vulnerability Related Demand

Examining the differences in vulnerability related demand, we find there is a 125x difference between the highest and lowest demand

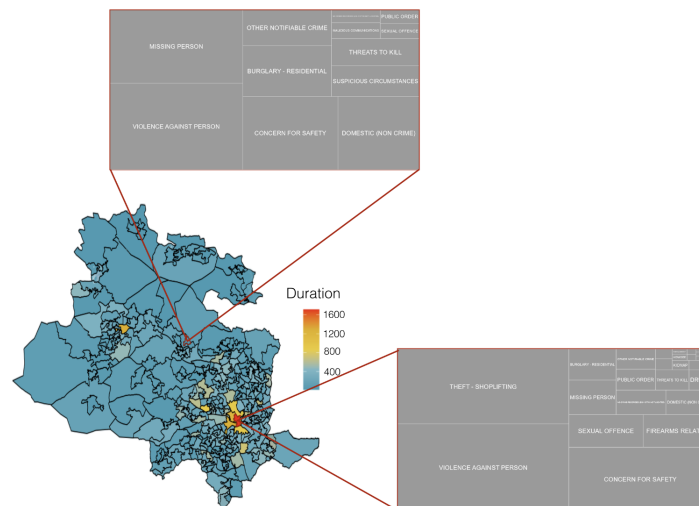


Figure 4: Spatial distribution of duration density across the Bradford Region. Incidents which were voted vulnerability-related by experts. With proportional area charts